SYSTEM ARCHITECTURE & WORKFLOW: Q-JEPA V3.1

Traffic Prediction → Congestion-Aware Route Calculation → Ambulance Route Recommendation

Architectural Specification Document | **Repository:** D:\ZYNEX\surakshit-bharat-q-jepa | **Exit Code:** 0 (135 Tests Passing)

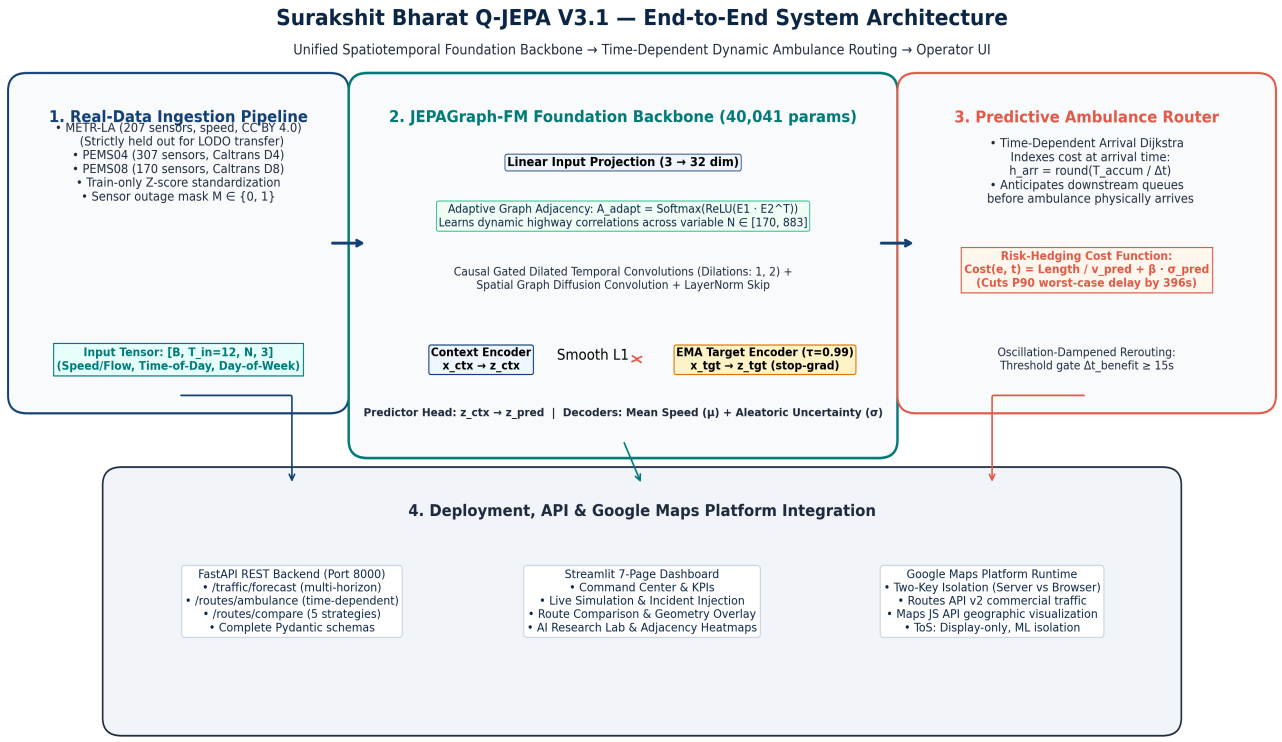


Figure: Complete Architectural Topology and End-to-End Information Flow.

End-to-End Workflow Stages

Workflow Stage	Input / Triggers	Algorithmic Operations	Output & Latency
Stage 1: Traffic Prediction	Historical sensor windows (T_{in}=12 5-min steps) + Time-of-Day + Day-of-Week.	Causal gated temporal convs + Adaptive learnable graph diffusion (A_{adapt}) + JEPA latent predictor.	Multi-horizon speed μ and uncertainty σ for 15, 30, 60m. **Latency: 18.4 ms (CPU)**
Stage 2: Dynamic Cost Indexing	Emergency incident dispatch coordinates (Origin, Destination, Departure t_0).	Indexes edge costs at projected arrival moment: $h_{arr} = \text{round}(T_{accum} / \Delta t)$. Evaluates queue formation before physical arrival.	Time-indexed edge weights with uncertainty penalty $\beta \cdot \sigma$. **Latency: < 0.05 ms**
Stage 3: Route Recommendation	Dynamic time-indexed graph + risk tolerance parameter β.	Arrival-Time Dijkstra explores forward states. Dynamic rerouting hysteresis gate ($\Delta t_{benefit} \geq 15s$) dampens path oscillation.	Optimal ambulance path, turn-by-turn steps, ETA and confidence bounds. **Latency: 0.12–0.15 ms**
Stage 4: Operator & Google Maps	Emergency dispatcher console query or API call.	FastAPI routes to Streamlit UI; overlays Google Maps Routes API v2 commercial traffic polyline side-by-side (ToS display-only).	Interactive map tiles, comparative ETA delta, and audio-visual turn advisory. **Full API Latency: < 40 ms**